Unit: 6 Unit 8: Quadratics **Name:** ________________
Topic: Systems of Quadratic Equations **Date:** ___________

Advanced (Carolina Reaper)

Q1. Solve the system: $x^2 + y^2 = 4$, $x^2 - y^2 = -1$
(A) $(-1, -sqrt2)$
(B) $(-1, sqrt2)$
(C) $(1, -sqrt2)$
(D) $(1, sqrt2)$
(E) No solution

Q2. What is the graphical representation of the system: $y = x^2$, $y = -x^2 + 2$?
(A) Two intersecting parabolas
(B) Two parallel parabolas
(C) One parabola and one line
(D) Two intersecting lines
(E) A circle and a parabola

Q3. Solve the system: $x^2 - 4y^2 = 1$, $x^2 + 4y^2 = 5$
(A) $(2, 0)$
(B) $(-2, 0)$
(C) $(1, 1)$
(D) $(-1, -1)$
(E) No solution

Q4. What is the solution to the system: $y = x^2 - 2$, $y = -x^2 + 2$?
(A) $(0, -2)$
(B) $(0, 2)$
(C) $(1, 0)$
(D) $(-1, 0)$
(E) No solution

Q5. Solve the system: $x^2 + y^2 = 9$, $y = x^2 - 4$
(A) $(-2, 0)$
(B) $(2, 0)$
(C) $(-1, -3)$
(D) $(1, 3)$
(E) No solution

Q6. What is the graphical representation of the system: $y = x^2 + 1$, $y = x^2 - 1$?
(A) Two intersecting parabolas
(B) Two parallel parabolas
(C) One parabola and one line
(D) Two intersecting lines
(E) A circle and a parabola

Q7. Solve the system: $x^2 - y^2 = 1$, $x^2 + y^2 = 2$
(A) $(1, 0)$
(B) $(-1, 0)$
(C) $(0, 1)$
(D) $(0, -1)$
(E) No solution

Q8. What is the solution to the system: $y = x^2 + 2$, $y = -x^2 - 2$?
(A) $(0, 2)$
(B) $(0, -2)$
(C) $(1, 0)$
(D) $(-1, 0)$
(E) No solution

Open Ended Questions (FRQ)

Q9. Solve the system: $x^2 + y^2 = 16$, $y = x^2 - 3$ and graph the solution set.

Q10. Find the solution to the system: $y = x^2 - 1$, $y = -x^2 + 1$ and describe the graphical representation of the solution set.

Chef's Tips

- Tip 1: Make sure to check for extraneous solutions.
- Tip 2: Graphing the solution set can help in identifying the correct answer.
- Tip 3: Consider using substitution or elimination methods to solve the system.